

CURRITUCK COUNTY REGIONAL, NC, USA

WMO: 723626

Lat: 36.399N

Lon: 76.016W

Elev: 5

StdP: 101.27

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 03737

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.8	-3.4	-16.1	0.9	-1.8	-13.1	1.2	-0.2	9.3	6.1	8.7	4.5	2.4	320	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.3	33.7	24.8	32.4	24.3	31.3	23.8	26.4	30.8	25.7	29.8	25.1	29.1	3.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	20.4	28.1	24.7	19.7	27.8	23.9	18.8	27.3	81.9	30.7	79.1	29.6	76.5	29.2	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.6	6.8	-8.9	36.2	3.4	1.1	-11.3	37.0	-13.3	37.7	-15.2	38.3	-17.6	39.1		
(5)			-10.1	27.3	3.0	1.1	-12.2	28.1	-14.0	28.7	-15.7	29.4	-17.8	30.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	5.7	6.9	10.1	15.7	19.9	24.4	26.1	25.8	22.5	17.3	11.3	8.3		
	DBStd	8.45	5.69	5.27	5.34	4.77	4.15	3.10	2.50	2.13	2.94	4.65	4.73	5.19		
	HDD10.0	484	156	115	69	7	0	0	0	0	0	4	38	95		
	HDD18.3	1719	391	320	261	107	33	1	1	0	3	76	214	312		
	CDD10.0	2754	24	29	71	178	306	432	498	489	375	231	78	43		
	CDD18.3	947	0	1	4	28	81	183	240	231	128	44	4	2		
	CDH23.3	7764	0	4	32	201	583	1586	2298	2031	798	216	11	4		
	CDH26.7	2525	0	0	3	43	167	574	846	672	184	37	1	0		
(14) Wind	WSAvg	2.7	2.9	3.1	3.4	3.4	2.8	2.4	2.0	1.9	2.5	2.4	2.5	2.6		
(15) Precipitation	PrecAvg	1237	94	75	97	84	96	116	144	139	125	98	78	88		
	PrecMax	1661	174	166	205	183	195	273	321	248	323	247	211	219		
	PrecMin	759	28	20	30	25	22	31	35	47	21	2	6	28		
	PrecStd	213	41	36	41	38	43	52	57	57	82	59	52	45		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.6	26.7	30.2	33.1	35.3	35.1	34.8	32.3	29.7	24.8	23.7		
		MCWB	17.0	17.2	17.1	20.0	22.0	23.8	25.6	25.6	23.7	22.4	19.7	19.6		
	2%	DB	19.0	20.8	23.7	27.8	30.6	33.0	33.7	32.7	30.6	27.6	22.5	21.0		
		MCWB	16.0	15.2	16.3	18.9	21.8	24.1	25.3	25.1	23.1	21.5	18.1	18.4		
	5%	DB	17.3	18.4	21.3	25.9	28.6	31.8	32.4	31.6	28.7	25.8	20.8	18.8		
		MCWB	14.2	14.3	15.2	18.1	21.2	23.3	24.8	24.5	22.7	20.8	16.9	16.8		
	10%	DB	15.0	16.0	18.9	23.5	26.7	30.0	31.1	30.2	27.4	24.0	18.9	17.1		
		MCWB	12.5	12.5	14.8	17.4	20.0	22.9	24.3	24.0	22.3	20.4	16.1	15.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	18.2	19.3	22.9	24.1	25.8	27.3	27.3	25.2	24.7	21.5	21.1		
		MCDB	19.7	21.7	23.8	26.0	29.0	30.7	32.3	32.0	28.8	27.2	22.9	22.2		
	2%	WB	16.6	16.9	18.0	20.9	23.1	25.2	26.4	26.5	24.6	23.6	19.5	18.9		
		MCDB	18.7	19.7	21.6	25.4	28.2	30.0	31.0	30.8	27.8	25.6	21.3	20.4		
	5%	WB	14.9	15.3	16.7	19.6	22.2	24.6	25.7	25.7	24.0	22.6	18.0	17.2		
		MCDB	16.8	17.8	19.9	23.6	26.4	29.2	30.1	29.6	27.1	24.6	19.8	18.5		
	10%	WB	12.9	12.7	15.2	18.4	21.4	23.9	25.2	25.2	23.4	21.2	16.5	15.2		
		MCDB	14.9	15.5	18.7	22.4	25.2	28.3	29.4	28.6	26.4	23.3	18.4	16.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.3	9.4	9.8	9.3	8.7	8.3	7.8	7.8	9.1	9.7	9.1		
		MCDBR	12.2	12.5	12.7	12.2	11.2	10.5	9.7	9.4	9.5	9.9	10.7	10.2		
	5% WB	MCWBR	9.4	8.3	7.0	5.5	4.5	3.9	3.5	3.3	4.0	4.6	6.9	7.9		
		MCWBR	11.5	11.9	10.9	10.2	9.5	9.2	8.8	8.3	7.5	7.3	9.1	9.6		
(40) Clear-Sky Solar Irradiance	taub		0.314	0.325	0.351	0.394	0.438	0.487	0.514	0.504	0.412	0.361	0.329	0.319		
	taud		2.535	2.497	2.454	2.344	2.254	2.168	2.104	2.121	2.376	2.505	2.542	2.550		
	Ebn at Noon		885	915	917	889	850	806	780	779	842	862	859	855		
	Edh at Noon		79	92	105	123	137	149	158	153	112	90	78	74		
(44) All-Sky Solar Radiation	RadAvg		2.36	3.13	4.20	5.37	6.03	6.40	6.09	5.49	4.51	3.61	2.67	2.12		
	RadStd		0.18	0.32	0.32	0.38	0.46	0.38	0.30	0.35	0.32	0.39	0.20	0.17		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+170	+112

Nomenclature: See separate page